

The Hong Kong Polytechnic University

Subject Description Form

Subject Code	AP40014
Subject Title	Imaging: Detector, Display and Processing
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Exclusion: AP30003 & AP40002
Objectives	The aim of this subject is to provide a fundamental understanding of the operation of photodetectors, imaging systems based on semiconductor devices, the principles of display devices and imaging processing, and to illustrate the application of various types of photodetectors, display technology and imaging analysis.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Outline and explain the basic principles of classical radiometry and blackbody radiation theory; (b) Describe and explain the physical characteristics of commonly used photodetectors and evaluate detector systems based on figures of merit, including signal to noise ratio, sensitivity; (c) Describe and discuss design issues in the development of an imaging system; explain the principle of representative imaging devices, including charge-coupled devices (CCD). (d) describe the human visual system and apply display specifications to understand the characteristics of display systems, explain the operation principles based on the fundamentals of materials science and fabrications of various display devices; analyze advantages and disadvantages of various display panels; (e) describe the structure of thin film transistors (TFT) and formulate the TFT characteristics, explain the operating principles of active matrix liquid crystal displays (AMLCD), semiconductor light emission diode (LED) and organic light emission diode (OLED) in displays; (f) explain representations and properties of imaging; and describe various methods for image processing and analysis.
Subject Synopsis/ Indicative Syllabus	Introduction: radiometry, blackbody radiation laws. Optical detectors: detection mechanisms (principles of thermal detectors and photon detectors, photoconductors, operation and characteristics of photo diodes, photomultiplier tube, microchannel plate.), introduction to noise and figures of merit of photodetectors; Imaging systems: charge-coupled devices (CCD), thermal imaging devices, films and digital photography, time and frequency analysis using Fourier transformation, sampling, filter and digitizing in imaging systems. image pre-processing; image enhancement, restoration and reconstruction; image segmentation; image understanding; automatic inspection and medical imaging applications.

	Fundamentals of display technology: human vision and perception; photometry and chromaticity; luminescence and phosphors for display; display specifications and measurements. Thin film transistors (TFT) for display: device structure and performance; amorphous and polycrystalline silicon TFTs. Liquid crystal display (LCD): LC phase, LCD structure; drive circuit; addressing method; fabrication of AMLCD. Light emission diode (LED) and organic light emission diode (OLED) displays: LED used for display; organic materials; device structure and performance; OLED display. Other types of display: plasma display panel (PDP); field emission display (FED); microdisplay; E-paper display and others. Image representation and models: continuous image function; image digitization; digital image properties; noise in images; data structures for image analysis.																																												
Teaching/Learning Methodology	The methodology includes classroom teaching and laboratory experiments. The teaching session will focus on basic concepts and principles of detector and imaging devices, the fundamentals of various display technologies and imaging processing, which are all related to the learning outcomes. The laboratory session will enhance the ability of the students in using various devices for light detection and displays.																																												
Assessment Methods in Alignment with Intended Learning Outcomes	<table border="1"> <thead> <tr> <th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr> </thead> <tbody> <tr> <td>1. Continuous Assessment</td><td>40</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>2. Examination</td><td>60</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr> <tr> <td>Total</td><td>100 %</td><td colspan="6"></td></tr> </tbody> </table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Continuous assessment consists of assignments, laboratory reports and mid-term test. The continuous assessment will assess the students' understanding of basic concepts and principles in materials science. Examination will be conducted to make a comprehensive assessment of students' intended learning outcomes as stated above.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Continuous Assessment	40	✓	✓	✓	✓	✓	✓	2. Examination	60	✓	✓	✓	✓	✓	✓	Total	100 %						
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Student Study Effort Expected	Class contact:																																												
	▪ Lecture & Tutorial						32 Hrs.																																						
	▪ Laboratory						9 Hrs.																																						

	Other student study effort:	
	▪ Self-study	79 Hrs.
	Total student study effort	120 Hrs.
Reading List and References	1. B E A Saleh and M C Teich, Fundamentals of Photonics, 2nd Edition, Wiley, 2007. 2. R W Boyd, Radiometry and the Detection of Optical Radiation, Wiley, 1983. 3. Jiun-Haw Lee, David N. Liu and Shin-Tson Wu, "Introduction to Flat Panel Displays", John Wiley & sons Ltd., 2008. 4. Ernst Lueder, "Liquid Crystal Displays: Addressing Schemes and Electro-Optical Effects", 2nd Edition, Wiley, 2010. 5. Milan Sonka, Vaclav Hlavac and Roger Boyle, "Image Processing, Analysis, and Machine Vision", 4th Edition, Global Engineering, 2013. 6. Rafael C. Gonzalez and Richard E. Woods, "Digital Image Processing", Pearson Education Limited 2018.	